

Minor- III: Real Analysis

College For Integrated Studies
University of Hyderabad

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Duration: 60 minutes
Maximum Score: 20 points

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Course Code: MM 263

Instructions: Answers without justification will receive a score of zero.

1. Let $f : [0, 1] \mapsto \mathbb{R}$ be continuous function on $[0, 1]$ and $f(0) < 0 < f(1)$. Then show that $\exists c \in (0, 1)$ such that $f(c) = 0$. [5]

2. Let $f : [0, 1] \mapsto \mathbb{R}$. Assume that $t_0 \in (0, 1)$ is a point of local minimum. If f is differentiable at t_0 then show that $f'(t_0) = 0$. [5]

3. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(t) = \begin{cases} e^{-\frac{1}{t^2}} & \text{if } t \neq 0, \\ 0 & \text{if } t = 0. \end{cases}$

Examine whether f is differentiable at 0. [5]

4. Let $f : (0, 1) \mapsto \mathbb{R}$ be differentiable such that $f'(t) = 0, \forall t \in (0, 1)$. Show that f is constant function on $(0, 1)$. [5]

All the best !